

Assignment 2. Deep Learning in Data Science.

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Exercise 1

For this assignment, I made the changes needed to the code of the previous assignment in order to adapt it to two hidden layers. To check the gradients, I adapted the functions based on 1) a slower but more accurate version based on the centered difference formula and 2) a faster but less accurate based on the finite difference method. Due to computational time, I tried with 5 hidden layers instead of 50, and the results were successful (table 1).

	Gradient 1	Gradient 2
W_0	1.0905e-08	9.8316e-07
W_1	1.1659e-08	2.1812e-06
b_0	1.4521e-08	1.3387e-06
b_1	1.4970e-08	1.7782e-06

Table 1: Relative difference of gradients.

I also new that the code was free of errors because of the trainings done with cyclic learning were. The replicates of the plot in assignment 2 are shown in figures 1,2.

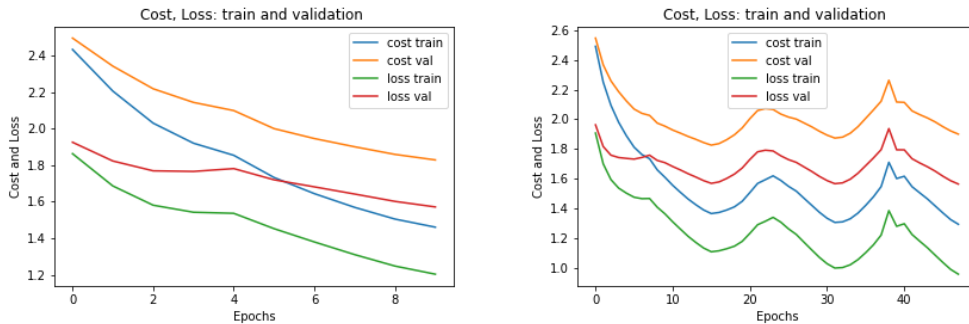


Figure 1: Cost and loss for one cycle of training. Figure 2: Cost and loss for three cycles of training.

As can be seen in the plots, when there is one cycle the cost first increases, as the learning

rate does, and then decreases, following the trend of the learning rate. When there are three cycles, the behaviour is similar, except for the values of the learning rate. The curves are very similar and repeated every cycle, but we can see that in every cycle the loss decreases. The accuracy achieved for one learning rate was 46.04%, and for three cycles 46.23%. Although there was no big improvement in this case by increasing the number of cycles, in general it brings benefits.

Coarse search

Initially, the set of values of lambda used for coarse search was $[0.056, 2.6e-05, 0.0015, 0.0021, 1.5e-05, 3e-05, 2e-05, 9.9e-05]$. Log scale was used with a minimum and maximum of $[-1, -5]$. The number of cycles was set to 2. The hyperparameter settings was always 40 epochs, 100 as batch size, eta min $1e-5$ and eta max 0.1. The best three lambdas were $[2e-5, 0.0021, 9.9e-05]$. For lambda $2e-5$ the validation accuracy was 50.2%, the highest.

Fine search

For this case, the set of values of lambda used for fine search was $[1e-05, 9e-06, 2.1e-05, 2e-06, 1e-06, 3e-06, 3.2e-05, 7.7e-05]$. Log scale was used with a minimum and maximum of $[-4, -6]$. The number of cycles was set to 2. The hyperparameter settings was always 40 epochs, 100 as batch size, eta min $1e-5$ and eta max 0.1. The best three lambdas were $[3e-06, 2.1e-05, 3.2e-05]$. For lambda $3e-06$ the validation accuracy was 51.7%. Since the accuracy obtained with lambda set to $2e-5$ is the highest, I performed a longer training with that setting.

For the last training, the total number of cycles was set to 5, the lambda used was $2e-5$ and the rest of the hyperparameters are the ones mentioned before. The accuracy in the validation set was 50.8%, and in the test set it was 51.4%. The accuracy is very similar to the ones already obtained, but a bit lower. This might be due to randomness or to the relation train data/value of lambda. The plots are shown in figures 3,4. The loss decreases with the number of epochs but the cost stays constant. For the accuracy, the validation accuracy is constant throughout the epochs, but the train accuracy increases, which might mean a bit of overfitting.

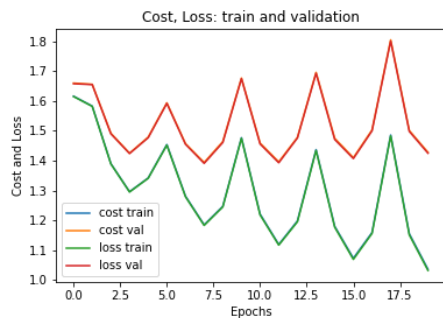


Figure 3: Cost and loss for validation and train data.

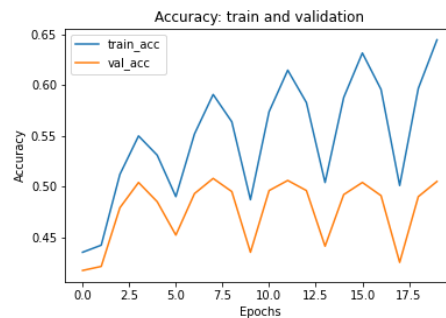


Figure 4: Accuracy for the validation and train data.

Bonus points

The network was optimized by changing several parts of the network. Several activation layers were implemented, such as ELU, leaky RELU or Swish. The best results were given by the original activation layer used, RELU, even though they were very similar for all the activation layer (it did not have a great impact). Then, a different regularization method was implemented, L_1 regularization, instead of L_2 regularization. Again, the results were very similar for both types of regularization, but slightly better for L_2 regularization. Finally, the number of hidden nodes was changed. A grid search was performed with different values of the hidden node, $[50, 250, 500]$ and different values of regularization, $[0.01, 0.001, 0.0001, 2e - 5]$. Since the model was robust regarding the change of activation layers and using different ind of regularization, only hidden nodes and regularization value were included in the grid search. In this way, a wider range of values could be included in the grid search. The best result was obtained for the configuration of the previous exercise, with mainly 250 hidden nodes, regularization $2e - 5$, and a reduction of η_{max} of 10% every cycle. In general the grid search was done for two cycles, but a final training of three cycles was done with the best configuration. The result can be seen in figure 8.

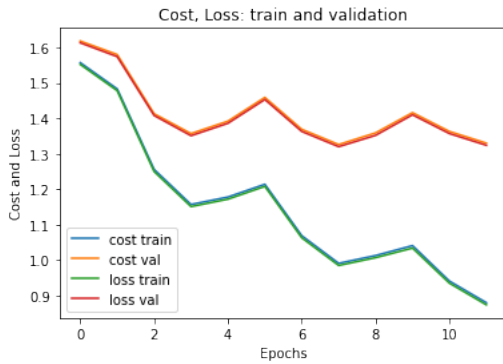


Figure 5: Cost and loss of the best configuration.

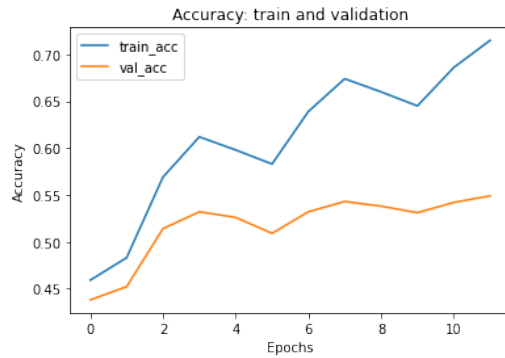


Figure 6: Accuracy of the best configuration.

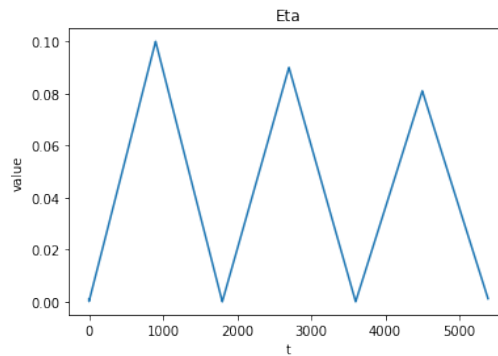


Figure 7: Evolution of the learning rate during the training of the best configuration. η_{max} was reduced every cycle.

Figure 8: Result of the best configuration

The best validation accuracy was 54.8%, and its correspondent test accuracy was 0.541%, which is a great improvement. The best improvements were accomplished by increasing the amount of training data and increasing the number of hidden nodes. Even though the accuracy has improved, there is overfitting, so different methods such as dropout should be applied. So far, by applying regularization, the gap is smaller but the model loses predicting power. Therefore a further search of values has to be done or other regularization methods have to be applied.